

The Geography of Everyday Gambling Exposure in London: Why London's poorest neighbourhoods are surrounded by betting shops?

Walk through parts of the city, and you may pass several bookmakers on a single high street. Walk through others, you will not see one for miles. This uneven landscape is not accidental. National research has already shown that Britain's poorest communities host far more betting shops than its wealthiest ones [1].

To understand how deeply this pattern runs through the capital, this article looks at gambling exposure neighbourhood by neighbourhood, using a simple yardstick: how many betting shops exist for every 10,000 people, roughly the size of a typical local community.

Key spatial patterns

1. Distribution of gambling exposure across London

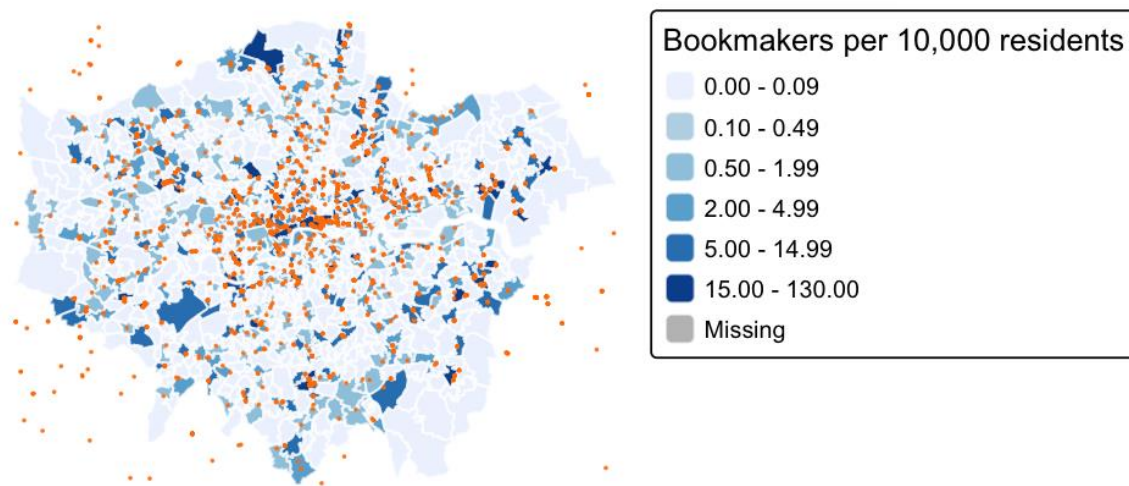


Figure 1: Bookmakers per 10,000 residents by London neighbourhood (darker shading = more bookmakers). Gambling shops cluster in inner-city areas and some East London boroughs, while many outer and well-off areas have none [2-3].

Gambling exposure is highly uneven across London. It clusters in neighbourhoods already under strain, revealing a clear pattern of place-based inequality. In the same areas where households face insecure work, crowded housing, and rising rents, residents are also far more likely to live among dense concentrations of betting shops.

Across most of the city, betting shops are rare or entirely absent. Yet in a small number of neighbourhoods, particularly in parts of central and East London, they line the same streets repeatedly. Of London's 1,002 neighbourhoods, more than half contain no betting shops at all [2-4]. At the other extreme, the most exposed areas, such as Westminster and the City of London, have over 110 bookmakers per 10,000 residents. This is not a gradual pattern but a sharp imbalance, illustrating how intensely gambling exposure is concentrated within a limited number of communities.

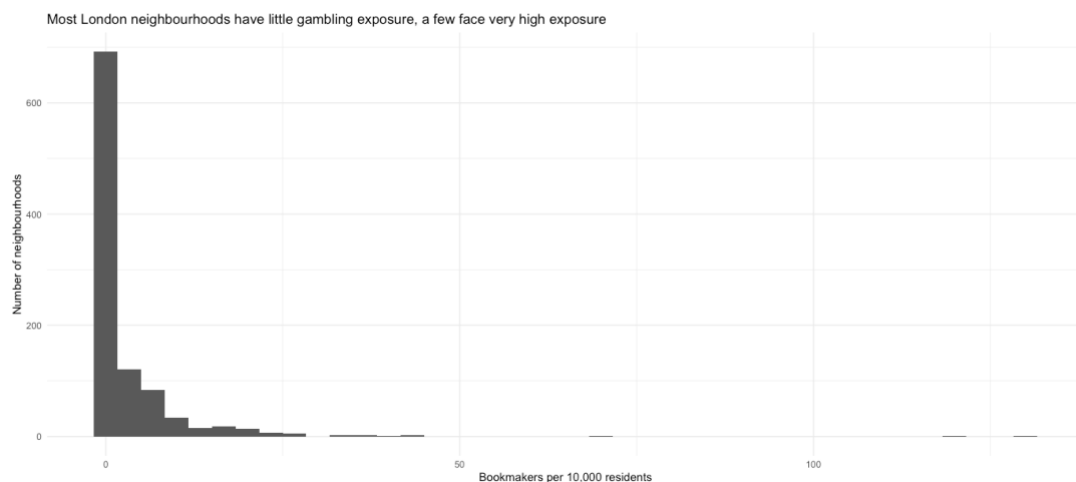


Figure 2: Histogram of bookmakers per 10,000 residents across London neighbourhoods. Most areas see little or no gambling exposure; a few face very high exposure [2-3].

2. Spatial patterns of gambling exposure and neighbourhood vulnerability

When we look more closely, this concentration follows familiar lines of social pressure. Figure 3 compares unemployment with gambling exposure, and a clear pattern begins to appear. Neighbourhoods with higher joblessness tend to have more betting shops on their streets. Areas with more overcrowded housing and higher levels of private renting also experience heavier gambling exposure. Put simply, many of the everyday signs of

disadvantage, from insecure work to cramped homes and rising rents, tend to travel together with a dense presence of betting shops.

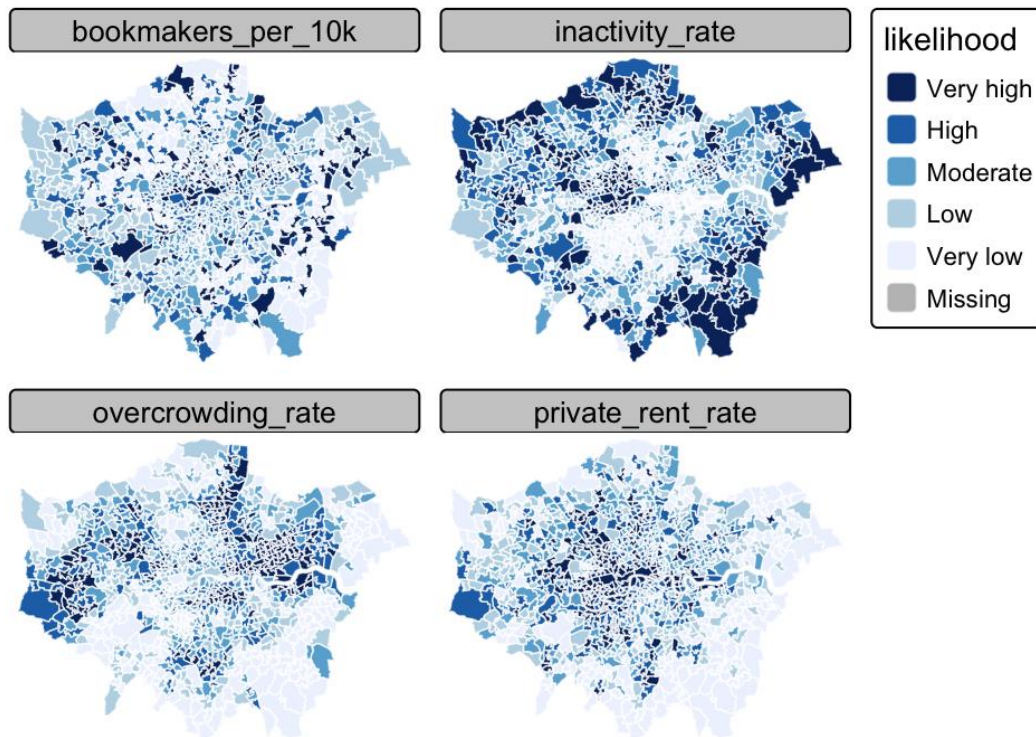


Figure 3: Neighbourhood gambling exposure and social vulnerability in London [2-4].

3. Explaining gambling exposure: regression results

To bring these patterns to life, we grouped neighbourhoods into five bands, from the least exposed to the most exposed to betting shops. Figures 4 compare these groups across a range of social measures. The message is remarkably consistent. Neighbourhoods with the heaviest concentration of betting shops also show the highest levels of unemployment, overcrowding, economic inactivity and private renting.

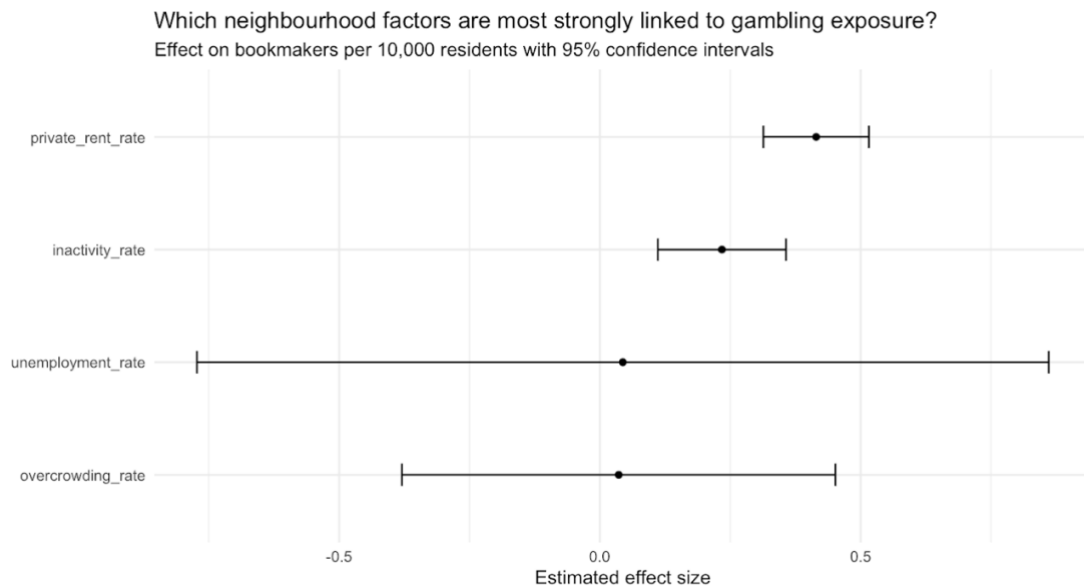


Figure 4: Which neighbourhood factors most strongly predict gambling exposure (bookmakers per 10,000 residents)? Each dot is the effect size of a 1-percentage-point increase in the factor, with 95% confidence bars. Higher private renting and higher inactivity show significant positive effects; unemployment's effect is negligible once other things are controlled for [2-4].

Take unemployment as one example. In the most exposed five neighbourhoods, joblessness is several percentage points higher than in the least exposed five. Across every measure, the same story repeats itself. In London, areas with many betting shops are very often the same places facing the greatest economic and housing pressures [1].

A 10% increase in private renting is associated with roughly four additional betting shops per 10,000 residents, even after accounting for unemployment, overcrowding, and economic inactivity. A similar increase in economic inactivity corresponds to about two extra betting shops per 10,000 people. Together, these conditions explain around 7% of the variation in gambling exposure across the city (see Table A2).

Overall, gambling outlets in London are not randomly distributed. They cluster in neighbourhoods already facing economic and housing pressure. This pattern is not unique to the capital. A national study reported by the Guardian found that the poorest 10% of areas across Britain contain over 20% of all gambling outlets, while the wealthiest 10% contain just 2% [1].

4. Neighbourhood clustering and spatial diagnostics

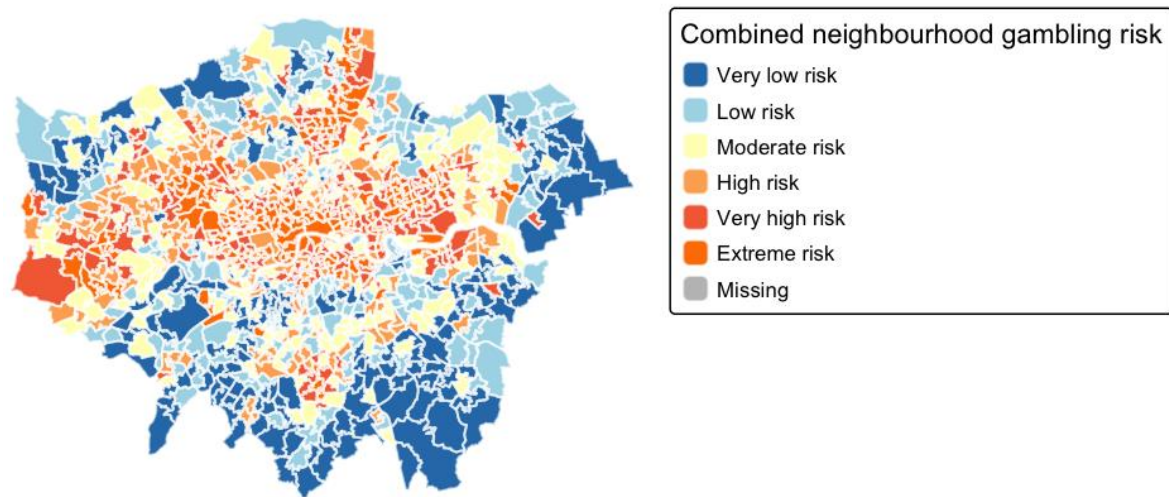


Figure 5: Mapping cumulative neighbourhood gambling risk in London. The map integrates gambling exposure with key social vulnerability indicators to highlight neighbourhoods facing the greatest overall risk. Blue areas indicate lower risk, while orange areas mark neighbourhoods experiencing extreme cumulative risk [2-4].

To untangle these factors, we ran a regression predicting bookmakers per 10,000 residents from all neighbourhood characteristics simultaneously (see Figure 5). After accounting for other influences, private renting is the strongest predictor of heavy gambling exposure. Neighbourhoods with higher proportions of private renters, often among the city's poorer areas, host markedly more betting shops. Economic inactivity also shows a clear positive association.

By contrast, once other conditions are held constant, unemployment itself has almost no additional effect, as its confidence interval spans zero. Overcrowding exhibits a smaller and less certain positive relationship. In simple terms, private renting is most strongly linked to where bookmakers concentrate.

The human consequences of this uneven geography are easy to recognise. In many lower-income neighbourhoods, betting shops are part of the everyday streetscape, while in wealthier parts of London, they may be almost invisible. During a recent parliamentary debate, MPs noted that in Brent, gambling shops now outnumber supermarkets in most wards [5]. One MP described walking along a deprived high street where “every second or third shop was a

gambling shop”, before visiting a wealthy street in Hampstead where there were none at all [5-6].

These patterns raise a clear policy dilemma. If poorer neighbourhoods host more betting shops, then gambling harm, which already falls disproportionately on vulnerable groups, becomes a question of inequality and fairness. Many local leaders have therefore called for stronger local control. Under current planning rules in England, councils are usually required to approve betting shops once a national licence is granted. MPs have argued that this should change, allowing communities to limit clusters of bookmakers in areas that are already struggling [5]. Reducing gambling harm is thus likely to require targeted local action, including stronger licensing powers and zoning rules, focused on the neighbourhoods facing the greatest risk.

Limitations and Conclusion

Gambling exposure reflects deeper urban inequalities linked to housing insecurity, labour-market exclusion, and uneven commercial investment. However, this analysis cannot capture every force shaping where betting shops appear. It relies on neighbourhood averages; the mapping may miss some venues, and the results represent a snapshot rather than long-term change. Commercial rents, planning decisions, footfall, and historical licensing patterns also influence outlet locations alongside social vulnerability.

Several central London neighbourhoods show exceptionally high exposure that is not fully explained by resident socio-economic conditions (Figure 1). As major tourism, employment, and transport hubs, these districts have daily populations far exceeding their residential base [1][5]. Gambling venues, therefore, serve largely transient populations, inflating per-capita exposure and introducing structural bias into residential-based models.

Even with these limits, gambling in London is not simply a private choice. It is shaped by place, housing conditions, and wider economic pressures, with betting shops concentrated in communities already under strain. Treating gambling as a form of spatial inequality, rather than solely a matter of individual responsibility, is essential if policy is to protect those most at risk.

Appendix

Data Sources and Preparation

Betting shop locations were obtained from OpenStreetMap (OSM) using all features tagged as gambling or bookmaker. Although OSM provides extensive geographic coverage, prior research indicates that gambling venues are systematically undercounted in volunteered geographic data [3]. Duplicate entries and non-London records were removed. All spatial datasets were projected to the British National Grid coordinate system to ensure accurate area measurement.

Neighbourhood geography was represented using the official 2021 Middle Layer Super Output Areas (MSOAs) for Greater London. MSOAs provide a stable spatial unit for socio-economic analysis, with an average population of approximately 10,000 residents.

Socioeconomic and demographic variables were sourced from the 2021 UK Census at the MSOA level. These included total population, unemployment rate, economic inactivity rate, overcrowding, private renting, and the proportion of residents aged 18–34.

Betting shop point data were assigned to MSOAs using a standard point-in-polygon spatial join procedure [6]. All datasets were subsequently merged by MSOA code to form the final analytical dataset, comprising 1,002 London neighbourhoods (MSOAs) and 2,767 betting shop locations.

Constructed Measures

Two exposure measures were constructed.

First, **betting shops per km²** was calculated by dividing the number of betting shops in each MSOA by its land area in square kilometres.

Second, **betting shops per 10,000 residents** was calculated as:

$$10,000 \times (\text{number of shops} \div \text{total MSOA population})$$

This population-normalised measure serves as the dependent variable in the regression analysis.

Neighbourhood vulnerability indicators were expressed as proportions and capture distinct dimensions of socio-economic and housing stress:

- Unemployment rate
- Economic inactivity rate
- Overcrowding rate
- Private renting rate
- Proportion of residents aged 18–34

Exploratory and Spatial Analysis

Spatial maps were used to visualise the distribution of gambling exposure and neighbourhood vulnerability across London. Descriptive statistics and distributional analysis were used to explore the data and inform model specification. Correlation checks indicated no problematic multicollinearity among predictors.

For visual presentation, several map legends were converted from strict quantile classifications to more interpretable categorical labels to improve readability for a general audience, while preserving the underlying data structure.

Table A1: Summary statistics for key variables

Variable	Mean	Median	Min	Max
Bookmakers per 10k	3.17	0.00	0.00	129.93
Unemployment rate (%)	3.87	3.85	1.42	7.23
Overcrowding rate (%)	4.20	4.12	0.44	9.44
Private renting rate (%)	11.72	10.99	1.78	34.17
Economic inactivity (%)	27.34	27.46	13.84	49.09

The final dataset covers 1,002 London neighbourhoods (MSOAs). Gambling exposure is highly uneven. 523 neighbourhoods (52%) contain no betting shops, while the most exposed area records 129.9 bookmakers per 10,000 residents. The distribution is therefore extremely right-skewed, with a small number of neighbourhoods carrying a disproportionate share of total exposure.

Regression Modelling

Several additional variables were examined during the initial analysis, including the proportion of residents aged 18–34, overall deprivation scores (IMD), and an alternative exposure measure based on betting shops per km². These variables were excluded from the final model because they either exhibited weak independent associations with gambling exposure or were highly correlated with the retained predictors. The final specification therefore prioritises interpretability and parsimony.

An ordinary least squares regression model was estimated:

$$\begin{aligned} \text{Bookmakers per } 10k_i &= \beta_i + \beta_1 \text{Unemployment}_i + \beta_2 \text{EconomicInactivity}_i \\ &+ \beta_3 \text{Overcrowding}_i + \beta_4 \text{PrivateRenting}_i + \varepsilon_i \end{aligned}$$

All predictors were entered simultaneously. Coefficients are interpreted as marginal associations. For example, a coefficient of 0.4 indicates that a one-percentage-point increase in the predictor corresponds to an average increase of 0.4 betting shops per 10,000 residents, holding other variables constant [7].

Table A2: Regression results predicting betting shops per 10,000 residents

Variable	Estimate	Std. Error	p-value
Unemployment rate	0.04	0.42	0.916
Overcrowding rate	0.04	0.21	0.865
Private renting rate	0.41	0.05	<0.001
Economic inactivity rate	0.23	0.06	<0.001

The model explains a modest but statistically significant share of the variation in gambling exposure ($R^2 = 0.068$).

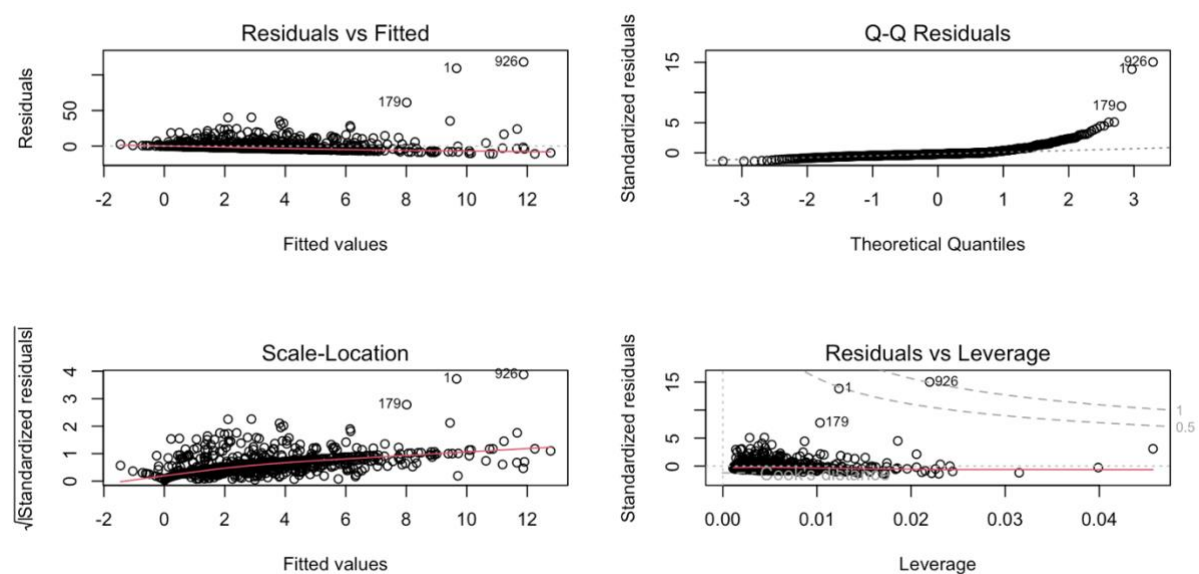
Given the right-skewed distribution of gambling exposure, log-transformed models were explored as robustness checks. These yielded substantively similar findings, and the linear specification was therefore retained to maximise interpretability and policy relevance.

The English Index of Multiple Deprivation (IMD 2019) was initially included but excluded from the final model due to strong correlation with private renting and economic inactivity, which reduced interpretability without improving explanatory power.

Model Evaluation

Model performance was assessed using the coefficient of determination (R^2), indicating moderate explanatory power typical of cross-sectional urban data [8]. Variance Inflation Factors confirmed the absence of problematic multicollinearity, with all values well below accepted thresholds [8]. Residual diagnostics revealed no major violations of linear model assumptions.

Figure A1: Regression diagnostic plots.



Residuals versus fitted values, Q–Q plot, scale–location plot, and residuals versus leverage for the final regression model, assessing linearity, normality, homoscedasticity, and the presence of influential observations.

Residuals exhibit no strong systematic pattern, supporting the linearity assumption. The Q–Q plot indicates moderate deviation from normality in the upper tail, reflecting a small number of high-exposure neighbourhoods. The scale–location plot suggests mild heteroscedasticity, a common feature of cross-sectional urban data, while the residuals versus leverage plot

identifies a small number of influential observations, none exceeding conventional thresholds.

Spatial autocorrelation in residuals was assessed using Moran's I. Only weak spatial dependence remained, indicating that the model adequately captured the principal spatial processes influencing gambling exposure [9]. The results are consistent with previous research showing that gambling venues disproportionately concentrate in socioeconomically deprived neighbourhoods [10].

Limitations

OpenStreetMap data are likely to undercount the true number of gambling venues [11], resulting in conservative exposure estimates. All results are ecological and cannot be interpreted at the individual level. The cross-sectional design precludes causal inference and limits insight into temporal dynamics. In central London districts, exceptionally high exposure partly reflects large transient and commuter populations, which inflate per-capita measures based on residential denominators. Although residual spatial dependence appears to be limited, more sophisticated spatial models could further refine the inference.

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